Physics Class 11 Federal Board
How to Smart Study in Next 2 Days. Best Tips to Score Maximum- Practicing Numerical and How to Attempt Them in Exams

i) Model Paper → AI, Gm, Cland

ii) Long Ker

Def/State

Math

units

Diagram

Derivation

Numerical → Formula

Given

To find

Numerical Short + Long

i) MCQs 6~8 17

ii)

ii) MCQs 17

Session → Long Ques

7/8

select

Ques

A turtle travel 45cm in 20 minutes, its average velocity will be:	2.25cm/h	90cm/h	135cm/h	15cm/h

$d = 45cm$, $t = 20min$

$60 min = 1 hr$

$1 min = \frac{1}{60} hr$

velocity = ?

$20 min = \frac{20}{60} hr$

$t = \frac{20}{60} hr = \frac{1}{3} hr$

$\vec{v} = \frac{45cm}{\frac{1}{3}} = .$

m/s

20 min $= 20 \times 60$ $= 1200$

1m = 100cm

$45cm = \frac{0.45m}{1200 s}$

$\boxed{v = \frac{d}{t}}$

$d = \lambda/2$ $\quad v = f\lambda$

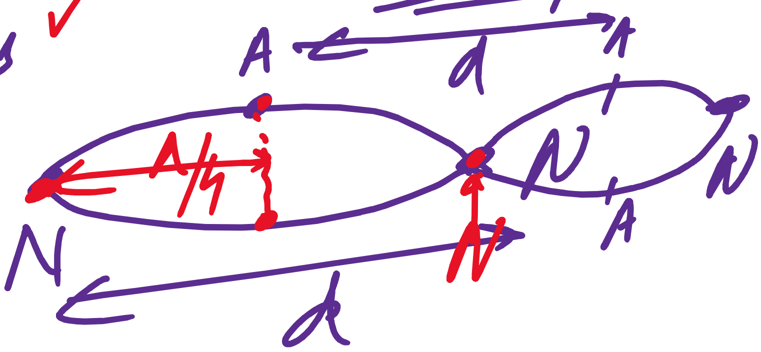
$\boxed{0.06}35$

A stationary wave is formed by waves of frequency 512Hz.The speed of the wave is 65m/s, the distance between two consecutive antinodes is:	0.126m	0.06m	0.03m	0.024

$f = 512\,Hz$

$v = 65\,m/s$

$d = \frac{\lambda}{2} \rightarrow (1)$



$\lambda = ?$

$N \rightarrow A$

$v, f, \lambda = ?$

$v = f\lambda$

$\lambda = \frac{v}{f} = \cancel{\frac{512}{65}} = \frac{65}{512} = 0.127\,m$

$d = \frac{\lambda}{2} = 0.0635\,m$

A wire is stretched to double of its original length, its strain is	0.5	1	zero	2

original length, its strain is	0.5	1	zero	2

$$\text{Strain} = \frac{\Delta L}{L_0} = \frac{L_0}{L_0} = 1$$

$$L \rightarrow 2L_0$$

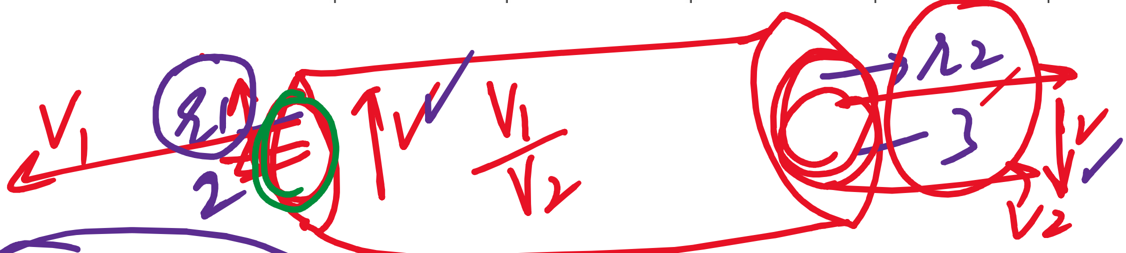
$$\Delta L, \quad \boxed{2L_0 - L_0} = L_0$$

$$L_0 - \frac{L_0}{2} = \frac{\frac{L_0}{2}}{L_0}$$

$$\frac{L_0}{2L_0} = \frac{1}{2}$$

$$A \propto \frac{1}{v}$$

The radius at two ends of a pipe is in the ratio of 2:3, then the speed of liquid at the two ends is in the ratio of:	2:3	3:2	4:9	9:4



$$\frac{r_1}{r_2} = \frac{2}{3}$$

$$A_1 V_1 = A V_2 = \text{const}$$

Area

speed

Area $\quad \propto$ speed

$AV \propto K \Rightarrow A = \frac{K}{V}$

$PV = C \quad V = \frac{K}{A} \quad \boxed{v \propto \frac{1}{A}}$

$P = \frac{C}{V}$

$\frac{r_1}{r_2} = \frac{2}{3} \rightarrow (i)$

$A_1 V_1 = A_2 V_2$

$\frac{V_1}{V_2} = ? \qquad \frac{V_1}{V_2} = \frac{A_2}{A_1} = \frac{\pi r_2^2}{\pi r_1^2}$

$\boxed{\frac{V_1}{V_2} = \frac{r_2^2}{r_1^2} = \frac{9}{4}}$

$\frac{r_1}{r_2} = \frac{2}{3} \Rightarrow \frac{r_1^2}{r_2^2} = \frac{4}{9} \Rightarrow \frac{r_2^2}{r_1^2} = \frac{9}{4}$

mgh

The force that acts on the body but it does no work is called:	Frictional force	Gravitational force	Centripetal force	Elastic force

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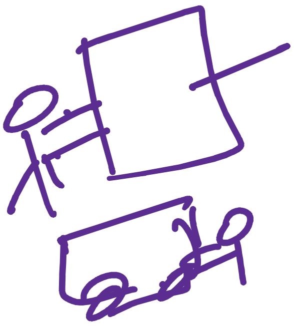
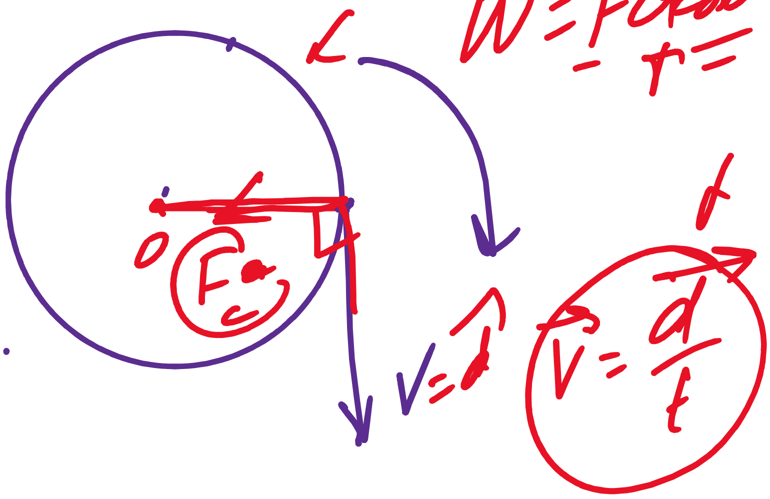
$W = Fd\cos\theta = Nm$



$W = zero$ $\theta = 90$ $\cos 90 = 0$

$F_c = \frac{mv^2}{r^2}$

$W = Fd\cos\theta$



$F_c d$

$v = \frac{d}{t}$

$9 + 16 = 25$

When Fx = 3 N and F = 5 N then Fy =	6N	8N	2N	4N

Par. / Rectan

$F^2 = F_x^2 + F_y^2$

$F_y^2 = F^2 - F_x^2$

$F = 5$ $F_x = 3N$ $F_y = ?$

3, 4, 5

$F_y = \ldots$ $F_x = 3N$

$= \sqrt{25} = 9$

$= \sqrt{16} = 4$

3, 4, 5

5, 12, 13

6, 8, 10

[illegible]

[illegible]

Find cross product of two unit vectors at 0° and 90°.	03

4/6

$$\vec{A} \times \vec{B} = A\,B\,\sin\theta$$

1) $\theta = 0°$ $\quad = 1 \cdot 1 \cdot \sin 0 = 0$

$\theta = 90°$ $\quad = 1 \cdot 1 \cdot \sin 90 = 1$

What is effect of radius of cross section of conductor on its resistance?

$$R = \rho \frac{L}{A}$$

$R \quad (A)$

$$\boxed{R \propto \frac{1}{A}}$$

$R = \rho \frac{l}{A}$, $\boxed{R \propto \frac{1}{A}}$

$r = 2m$,
$A = \pi r^2$

$\Rightarrow \boxed{R \propto \frac{1}{r^2}}$

i) If radius of cross-section increase, then Resistance will decrease

Radius of sphere is measured with a screw gauge and is substituted in the formulae of its surface area($A=\pi r^2$) and volume($V=4/3\pi r^3$). Which of the two results will be more accurate?

$\frac{\Delta r}{r}$

$2\pi r$ r

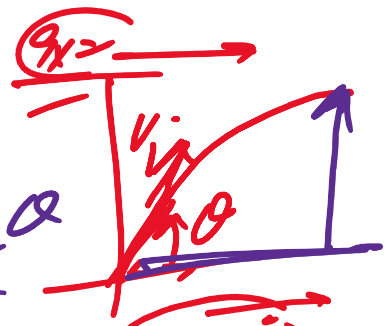
As surface Area depends on r^2
And Volume " " " r^3

So, Relative Error in Surface Area will be multiplied by 2

But in Relative Error in Volume will be $\times$ by 3 wrt x x by ?

Thus, Surface Area is more accurate.

Find the angle of projection of a projectile for which the maximum height and corresponding range are equal?



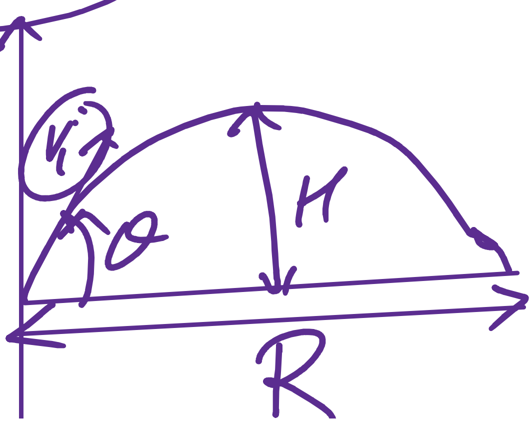
Given $H = R$

$$\frac{\cancel{v_i^2}\sin^2\theta}{2\cancel{g}} = \frac{\cancel{v_i^2}\sin 2\theta}{\cancel{g}}$$

$$\frac{\sin^{\cancel{2}}\theta}{2} = 2\cancel{\sin\theta}\cos\theta$$

$$\sin\theta = 4\cos\theta$$

$$\frac{\sin\theta}{\cos\theta} = 4 \Rightarrow \tan\theta = 4$$



$$\theta = \tan^{-1}(4)$$

$$\theta \approx 76^\circ$$

Q No 76.

R

A 40 kg mass is supported by 5 m long aluminium wire ($Y_A=7\times10^{10}$ Pa). To have same elongation in a copper wire of the same length under the same weight, find the diameter of copper wire. ($Y_C=12\times10^{10}$ Pa)

Long Question

$\boxed{F = W, L, \Delta L}$

$D_{cu} \Rightarrow d_{cu}$??

Cu

40 kg

ΔL

Y_A

Al

ΔL

Formula

$$\text{Stress} = \frac{F}{A}, \quad \text{Strain} = \frac{\Delta L}{L}$$

$$Y = \frac{F/A}{\Delta L/L}$$

$$\frac{F}{A} \div \frac{\Delta L}{L}$$

$$\frac{F}{A} \times \frac{L}{\Delta L}$$

$$Y = \frac{FL}{A\Delta L}$$

As F & L are same for both Cu + Al

$$\Rightarrow Y_{cu} = \frac{FL}{A_{cu}\Delta L}, \quad Y_{Al} = \frac{FL}{A_{Al}\Delta L}$$

$$Y_{Cu} = \frac{FL}{A_{Cu}\Delta L} \rightarrow (1) \qquad Y_{Al} = \frac{FL}{A_{Al}\Delta L} \rightarrow (2)$$

$$(1) \div (2) \Rightarrow \frac{Y_{Cu}}{Y_{Al}} = \frac{\cancel{FL}}{A_{Cu}\cancel{\Delta L}} \times \frac{A_{Al}\cancel{\Delta L}}{\cancel{FL}}$$

$$l \Rightarrow \frac{Y_{Cu}}{Y_{Al}} = \frac{A_{Al}}{A_{Cu}}$$

$$A_{Cu} Y_{Cu} = A_{Al} Y_{Al}$$

$$A_{Cu} = \frac{A_{Al} Y_{Al}}{Y_{Cu}}$$

c, l

$$A_{Cu} = A_{Al} \times \frac{7 \times \cancel{10^{10}}}{12 \times \cancel{10^{10}}} \qquad r = \frac{d}{2}$$

$$A = \pi r^2 = \pi \frac{d^2}{4}$$

$$\frac{\cancel{\pi} d_{Cu}^2}{\cancel{4}} = \frac{\cancel{\pi} d_{Al}^2}{\cancel{4}} \left(\frac{7}{12}\right)$$

$$\sqrt{d_{Cu}^2} = \sqrt{d_{Al}^2 \; \frac{7}{12}}$$

$x = 1$

$y = 0.7x$

$$\boxed{d_{cu} = \sqrt{\frac{7}{12}} \times d_{Al}}$$

$$\boxed{d_{cu} = 0.76\, d_{Al}}$$

An ideal liquid is flowing through a pipe of variable diameter as shown in figure. Where is pressure high (at position A or at B). Justify your answer.

Solution

By using Eqn. of continuity

$AV_1 = AV_2 \Rightarrow$ speed $\propto \frac{1}{Area}$

As it is obvious that Area of position A is larger compared to position B.

i.e. $Area_A > Area_B$

$\Rightarrow v_A < v_B$

Now, according to Bernoulli's Principle

$P \propto \frac{1}{\;}$

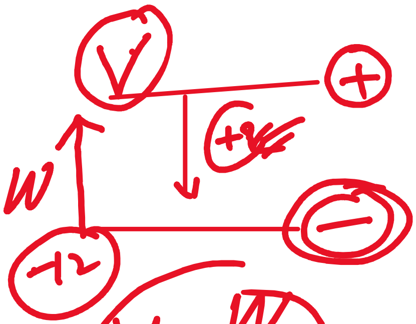
$$P \propto \frac{1}{v}$$

Since speed is less at A ~~Point~~ position, hence pressure at A will be higher

1C, 2C....

$$\boxed{P_A > P_B}$$

If 280 J of work is done in carrying a charge from a place where potential is -12V to another place where potential is V volt. Calculate the value of V.



$$V = \frac{W}{q}$$

$$\Delta V = \frac{\Delta W}{q}$$

$$\Delta W = q\,\Delta V$$

$$= q(V - (-12))$$

$$\Delta W = q(V + 12)$$

$$q = 1C$$

$$280 = 1(V + 12)$$

$280 = I(V + 12)$

$V = 280 - 12$

$\boxed{V = 268 \text{ volts}}$

Two spheres A and B are of the same masses but the radius of A is greater than that of B. Which one will require more torque applied on it to bring into rotation about their axes of rotation?

N

$\vec{\tau}$

$r_A > r_B$ ✓

Sol

$\tau = I\alpha \longrightarrow (1)$

I = moment of inertia

$\frac{2}{5}mr^2$

$\frac{1}{2}mr^2$

$I \propto mr^2$

$\Rightarrow I \propto r^2$

$I_A \propto r_A^2$

$I_A \propto r_A^2$

$I_B \propto r_B^2$.

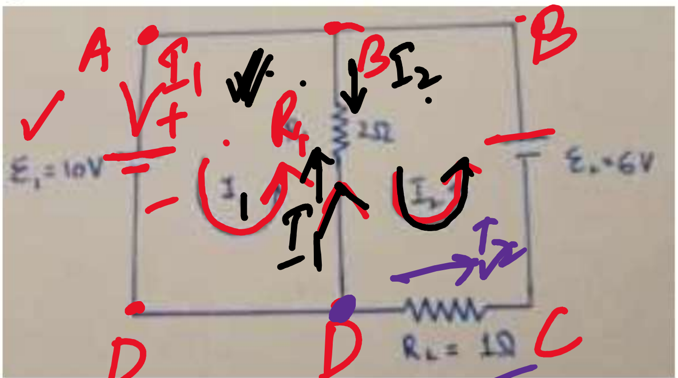
As $r_A > r_B \Rightarrow I_A > I_B$ ✓

$\Rightarrow \tau_A > \tau_B$ ✓

Hence more Torque is required in case of Sphere A

$V = -IR_1 = -R_1(I_1 - I_2)$

Find current flowing through each resistor, using Kirchhoff's law in the given circuit.



$\mathcal{E}_1$ $\mathcal{E}_2$

Mesh

I_1 I_2

$\sum V = 0$

Sol. In Loop 1 (ABDA)

$V =$

Voltage

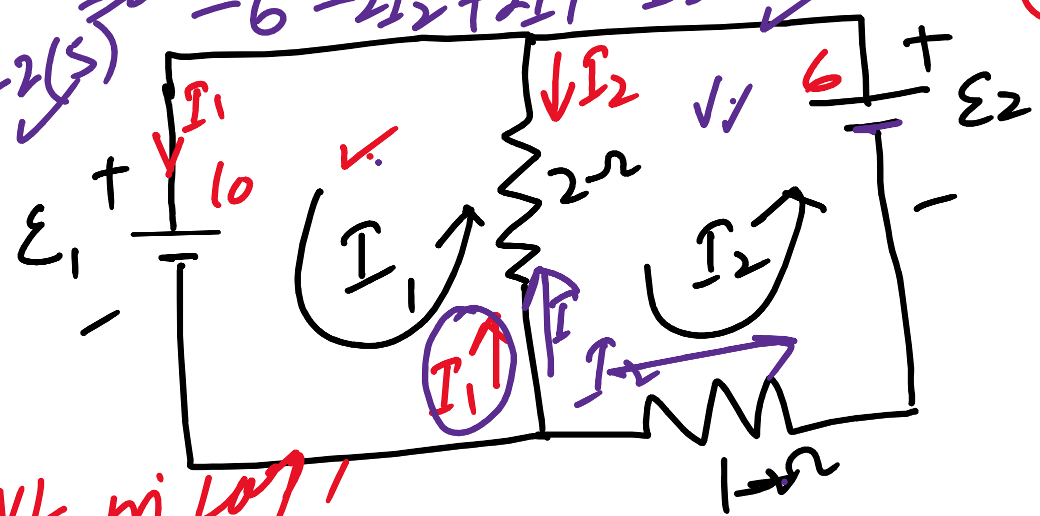
$+\mathcal{E}_1 -$

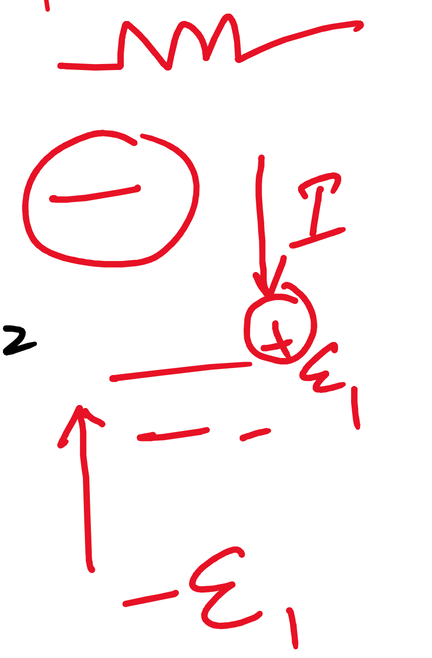
$\{-V\}$ Voltage Drop

Loop 2 $\quad -6-2(I_2-I_1)-I_2=0$

$-6-2I_2+2I_1-I_2=0$

$10-2(5)=0$





$9-4=5$

KVL in Loop 1

$10-2(I_1-I_2)=0$

$5-(I_1-I_2)=0$

$5-I_1+I_2=0$

$\Rightarrow I_1-I_2=5 \rightarrow (1)$

Loop 2 $\quad 2I_1-3I_2=6 \rightarrow (2)$

$\boxed{I_1=9A}$

$\boxed{I_1 = 9A}$ ✓

$\boxed{I_2 = 4A}$ ✓

Current through 2Ω Resis

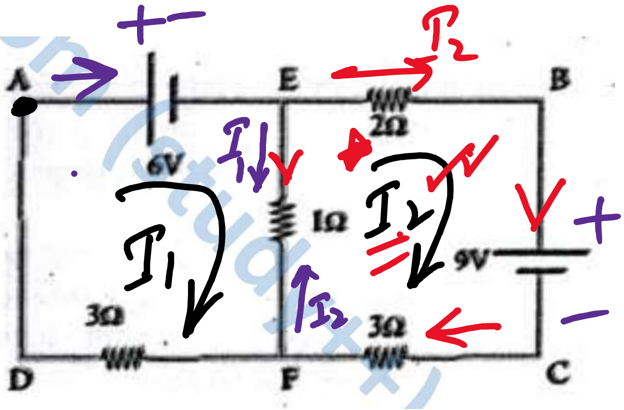
$I_1 - I_2 = 9 - 4 = 5A$

Current through 1Ω

$\boxed{I_2 = 4A}$

$V = -IR$

11.6 Calculate the current that flows in the 1Ω resistor in the following circuit. (Ans:0.13 A)



KVL in left loop (AEFDA)

$6 - 3I_1 - 1(I_1 - I_2) = 0$

$6 - 3I_1 - I_1 + I_2 = 0$

$6 - 4I_1 + I_2 = 0$

A=B
B=A

$$6 - 4I_1 + I_2 = 0$$

$B = A$

$$\Rightarrow \quad 4I_1 - I_2 = 6 \rightarrow (1)$$

KVL in Loop 2 (EBCFE)

$$-3I_2 - 1(I_2 - I_1) - 2I_2 + 9 = 0$$

$$-3I_2 - I_2 + I_1 - 2I_2 + 9 = 0$$

$$I_1 - 6I_2 = -9 \rightarrow (2)$$

$$\boxed{I_1 = \frac{45}{23} A}$$

$$\boxed{I_2 = \frac{42}{23} A}$$

Thus current through 1Ω resistor is

$$= I_1 - I_2$$

$$I = \left(\frac{45}{23} - \frac{42}{23}\right) A$$

$$I = \frac{3}{23} A$$

In an experiment a 3kg water rocket is launched from ground. The rocket total energy at the top of its flight is 2352J. a. What was the rocket launching speed? b. What height did the rocket reach? c. What is the kinetic energy and potential energy of the rocket after 2.5s of its launch?	02+02 +02

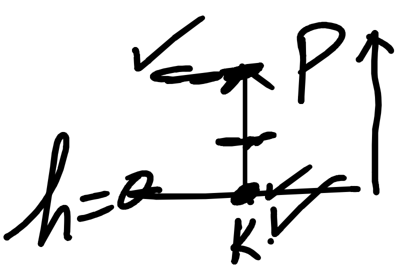
$m = 3kg$

$E_{Total} = 2352J$

$v = ?$

$K_E + P_E$

a) Sol = $E_{Total} = E_K + E_P$



Initially, $P.E = 0$

$\Rightarrow K.E = E_{Total}$

$$\frac{1}{2} m v_i^2 = 2352$$

$$v_i^2 = \frac{2 \times 2352}{m}$$

$W = Fd$

(Nm)

$$\sqrt{v_i^2} = \sqrt{\frac{2 \times 2352 J}{3 \text{ kg}}}$$

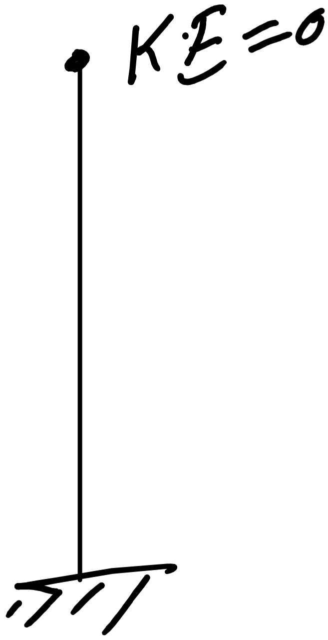
$$\boxed{v_1 = 39.6 \, m/s}$$

b) $E_{Total} = P.E.$

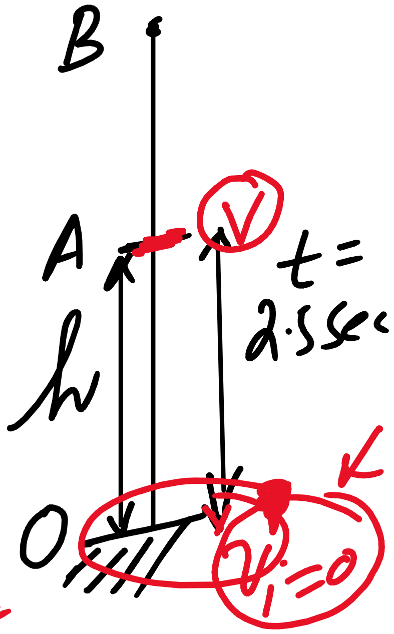
$$2352 = mgh_{max}$$

$$h_{max} = \frac{2352 J}{3 kg \times 9.8 m/s^2}$$

$$\boxed{h_{max} = 80 m}$$



c) $K.E = \frac{1}{2}mv^2 \rightarrow (1)$



$V_f = V_i - gt$

$V = v_i - gt$

$v = 39.6 - 9.8 \times 2.5$

$v = 15.1\ m/s$

$K.E_{2.5sec} = \frac{1}{2} \times 3kg \times (15.1)^2\ J$

$t = 2.5\ sec$

$P.E = mgh \rightarrow (2)$

$$\boxed{h = v_i t - \frac{1}{2}gt^2}$$

$(39.6)(2.5)$

$h =$

$(39.6)(2.5)/h$ ...

Estimate → Formula

$F_D = W = mg$